Flex 2.0

Use Case Descriptions

Total Use Cases: 26 | Actors: User, Student, Teacher, Admin, Secretariat, Saqib Bhai

UC01 Log In	
Environment	Flex 2.0 web portal, accessible via any modern browser (Chrome, Firefox, Edge). Available on both desktop and mobile.
Level	*User Goal*
Primary Actor	User (Student, Teacher, Admin, Secretariat, Saqib Bhai)
Stakeholders	• User — gains authenticated access to role-specific dashboard and features. • IT Security — ensures only authorised personnel access the system; prevents unauthorised entry. • University Administration — maintains an audit trail of login activity.
Precondition	• The user has a registered account in the system. • The system is online and reachable. • The user's account is not locked or suspended.
Guarantee	• Success: User is authenticated, a session token is created, and the user is redirected to their role-based dashboard. • Failure: No session is created; the system remains in its prior state.
Success Case	The user provides valid credentials and successfully accesses their personalised dashboard.
Trigger	User navigates to the login page and submits their credentials.
Sequence of Steps	1. User opens the Flex 2.0 login page. 2. System displays the login form (username/email + password fields). 3. User enters their credentials and clicks 'Log In'. 4. System validates the credentials against the user database. 5. System generates a session token (JWT) tied to the user's role. 6. System logs the login timestamp and IP address in the audit log. 7. System redirects the user to their role-specific dashboard.

Extensions	• 3a. Invalid credentials: System displays 'Invalid username or password'. Step count increments. • 3b. Five consecutive failures: Account is temporarily locked; system displays lockout message and sends unlock instructions via email. • 3c. User clicks 'Forgot Password': System redirects to the password reset flow. • 4a. Database unreachable: System displays a service-unavailable error; no session is created.
Technology	• HTTPS (TLS 1.2+) for all data in transit. • JWT (JSON Web Tokens) for session management with role-encoded claims.

UC02 Change Password	
Environment	Flex 2.0 web portal, Profile / Account Settings section. Accessible post-login to all authenticated users.
Level	*User Goal*
Primary Actor	User (Student, Teacher, Admin, Secretariat, Saqib Bhai)
Stakeholders	• User — wishes to update their password for security or personal preference. • IT Security — enforces password complexity policies and detects suspicious changes. • University Administration — ensures password policy compliance across all accounts.
Precondition	• User is logged in with an active session. • The user knows their current password.
Guarantee	• Success: Password is updated in the database; all other active sessions are invalidated; confirmation email sent. • Failure: Password remains unchanged.
Success Case	User successfully replaces their old password with a new, policy-compliant password.
Trigger	User navigates to Account Settings and selects 'Change Password'.
Sequence of Steps	1. User opens the Account Settings page. 2. System displays the Change Password form (current password, new password, confirm new password). 3. User enters their current password. 4. User enters and confirms the new password. 5. System verifies the current password against the stored hash. 6. System validates the new password against complexity rules (min. 8 characters, uppercase, number, special character).

	7. System hashes the new password and updates the database record. 8. System invalidates all active sessions except the current one. 9. System sends a confirmation email to the user's registered address.
Extensions	• 3a. Current password is incorrect: System displays an error; change is aborted. • 4a. New passwords do not match: System displays 'Passwords do not match'. • 4b. New password fails complexity rules: System lists the specific unmet requirements. • 4c. New password is the same as the current password: System rejects and prompts for a different password. • 9a. Email delivery fails: Password is still updated; a warning is logged for the IT team.
Technology	• HTTPS (TLS 1.2+). • bcrypt for password hashing (minimum cost factor 12). • JWT session invalidation on password change.

UC03 View Profile	
Environment	Flex 2.0 web portal, Profile section. Available to all authenticated users. Read-only view of personal account information.
Level	*User Goal*
Primary Actor	User (Student, Teacher, Admin, Secretariat, Saqib Bhai)
Stakeholders	• User — reviews personal information for accuracy and completeness. • Admin — responsible for the accuracy of the data displayed; may need to update it on request.
Precondition	• User is logged in with an active session.

	• A user record exists in the database.
Guarantee	• Success: Profile information is displayed; no data is modified. • Failure: An error message is shown; no data is modified.
Success Case	User views their complete profile details including personal information, role, and department.
Trigger	User clicks 'Profile' or 'My Account' from the navigation menu.
Sequence of Steps	1. User clicks on the Profile link in the navigation menu. 2. System retrieves the user's record from the database using the session token. 3. System renders the profile page displaying: Full Name, University ID, Email Address, Role, Department/Faculty, Contact Number, and Registration Date. 4. User reviews the displayed information.
Extensions	• 2a. Database error: System displays a generic 'Unable to load profile' error; IT is notified via internal logging. • 3a. Some fields are empty: System displays 'Not provided' placeholders for missing optional fields. • 3b. User notices an incorrect field: System provides a link to 'Request a Change' (directing to UC11).
Technology	• HTTPS (TLS 1.2+). • REST API GET endpoint for user profile data. • Role-based access control (RBAC) — users can only view their own profile. • Relational database (Oracle SQL server).

UC04 View Info

Environment	Flex 2.0 web portal, Information / Announcements section. Available to all authenticated users, with content filtered by role.
Level	*User Goal*
Primary Actor	User (every one, each having their own abstract view of things relevant to them)
Stakeholders	• Teacher — course details, departmental notices, and academic calendars. • Student — marks PLO report, Tentative Study Plan, Grade Report. • Admin — publishes and maintains the information displayed.
Precondition	• User is logged in. • At least one information record exists relevant to the user's role.
Guarantee	• Success: Relevant information is displayed; no data is modified. • Failure: An error or empty-state message is displayed.
Success Case	User views up-to-date academic and institutional information relevant to their role.
Trigger	User selects 'View Info' from the navigation menu.
Sequence of Steps	1. User navigates to the Information section. 2. System determines the user's role from the session token. 3. System queries the database for information records tagged to that role. 4. System displays the information, grouped by category (e.g., Timetables, Exams, Announcements). 5. User browses or searches for specific information.
Extensions	• 3a. No records found for the user's role: System displays 'No information available at this time'. • 3b. Database error: System displays a service error; IT is notified.

	• 5a. User searches with no matching results: System displays 'No results match your query'.
Technology	• HTTPS (TLS 1.2+). • REST API GET endpoint with role-based content filtering. • Full-text search capability (Elasticsearch or DB full-text index). • Relational database for information records.

UC05 Register Course	
Environment	Flex 2.0 web portal, Course Registration module. Active only during the designated registration window set by the university calendar.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — enrolls in desired courses to fulfil academic requirements. • Teacher — needs an accurate list of enrolled students for course planning. • Admin — manages enrollment limits, course availability, and registration windows. • Finance Department — needs fee records generated per enrollment (UC07). • Academic Office — ensures courses are taken in the correct academic sequence (UC06).
Precondition	• Student is logged in. • The registration window is currently open. • The student has no registration holds or financial blocks on their account. • The target course has available seats.
Guarantee	• Success: Student is enrolled in the course; a fee record is generated; enrollment count is incremented. • Failure: No enrollment record is created; no fee is generated.
Success Case	Student selects a course, passes prerequisite and availability checks, and is successfully enrolled.
Trigger	Student navigates to the Course Registration page and submits a registration request.
Sequence of Steps	1. Student navigates to the Course Registration module.

	2. System displays available courses filtered by the student's programme and semester. 3. Student selects a desired course and clicks 'Register'. 4. System invokes Verify Pre-requisites (UC06); result is returned. 5. System checks that at least one seat is available in the course. 6. System creates an enrollment record linking student to the course. 7. System invokes Generate Fee (UC07); fee record is created. 8. System confirms enrollment and sends a notification email to the student.
Extensions	• 4a. Prerequisites not met: System displays which prerequisites are missing; registration is rejected. • 5a. Course is full: System offers to add student to the waitlist. • 5b. Student is already registered in the course: System notifies 'Already enrolled' and aborts. • 7a. Fee generation fails: Enrollment is held in a pending state; Finance team is alerted. • 8a. Email notification fails: Enrollment is still saved; failure is logged.
Technology	• HTTPS (TLS 1.2+). • REST API with database transactions (ACID compliance to prevent duplicate enrollments). • Relational database with row-level locking for seat count updates. • SMTP email service for confirmation notifications.

UC06 Verify Pre-requisites	
Environment	Flex 2.0 back-end service layer. An internal sub-function invoked automatically during course registration. Not directly accessible by any user.

Level	*Sub-function*
Primary Actor	System (invoked by Register Course — UC05)
Stakeholders	• Student — must have completed required prior courses to enroll. • Academic Office — enforces the academic progression policy. • System — must reliably gate registration based on policy rules.
Precondition	• Triggered as part of Register Course (UC05). • Student ID and target Course ID are available. • The student's academic transcript is accessible in the database.
Guarantee	• Success: Returns PASS — all prerequisites are satisfied. • Failure: Returns FAIL — with a list of unmet prerequisite courses.
Success Case	System confirms that the student has completed all required prerequisite courses with passing grades.
Trigger	Invoked via «include» by Register Course (UC05) immediately after course selection.
Sequence of Steps	1. System retrieves the prerequisite course list for the selected course from the academic catalogue. 2. System retrieves the student's completed courses and grades from the transcript database. 3. System compares each prerequisite against the student's academic history. 4. System checks that each prerequisite was passed (grade meets minimum threshold). 5. System returns PASS or FAIL with details to the calling use case (UC05).
Extensions	• 1a. Course has no prerequisites: System immediately returns PASS. • 2a. Transcript data is unavailable: System returns an error to UC05; registration is blocked with a message to contact the Registrar. • 4a. Prerequisite was completed at another institution (transfer student): System flags for manual review by the Academic Office.

Technology	• Internal REST API microservice or database stored procedure. • Academic records database (read-only access from this sub-function). • Configurable grade-threshold rules per course.

UC07 Generate Fee	
Environment	Flex 2.0 Finance module, back-end service layer. An internal sub-function invoked automatically upon successful course registration. Not directly triggered by the student.
Level	*Sub-function*
Primary Actor	System (invoked by Register Course — UC05)
Stakeholders	• Student — receives a fee invoice and is liable for payment. • Finance Department — requires an accurate fee record for billing and reporting. • Admin — oversees fee generation and may apply manual adjustments (scholarship, waiver).
Precondition	• Triggered as part of Register Course (UC05) after enrollment record is created. • The student's enrollment record and course fee structure exist.
Guarantee	• Success: A fee record is created, linked to the student and course, and an invoice is accessible. • Failure: Fee record is not created; the enrollment is placed in a pending-payment state.
Success Case	System calculates the applicable fee, creates a fee record, and generates an invoice for the student.
Trigger	Invoked via «include» by Register Course (UC05) immediately after enrollment record creation.
Sequence of Steps	1. System retrieves the fee structure for the registered course (credit hours × fee rate). 2. System checks the student's account for any applicable scholarships, waivers, or discounts. 3. System calculates the final payable amount. 4. System creates a fee record in the Finance database, linked to the student ID, course ID, and enrollment date.

	5. System generates a PDF invoice document and stores it in the document management system. 6. System sends a fee notification email with the invoice to the student.
Extensions	• 2a. Scholarship data is unavailable: System proceeds with the full fee and logs the issue for Finance to review. • 4a. Fee record creation fails: System rolls back the enrollment; student is shown an error and advised to retry or contact Finance. • 6a. Email delivery fails: Fee record and invoice are still saved; failure is logged.
Technology	• Finance module REST API. • SMTP email service. • Relational database for fee records.

UC08 Submit Course Feedback	
Environment	Flex 2.0 web portal, Feedback module. Accessible to enrolled students during the designated feedback window. Responses are stored anonymously initially.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — provides honest feedback to contribute to course quality improvement. • Teacher — receives anonymised aggregated feedback to improve teaching. • Admin — monitors course and faculty quality through feedback reports. • Academic Office — uses data for accreditation and quality assurance.
Precondition	• Student is logged in.

	• Student is or was enrolled in the course being reviewed. • The feedback submission window is currently open. • Student has not already submitted feedback for this course in this period.
Guarantee	• Success: Feedback is saved anonymously; submission is marked as complete for this student/course pair. • Failure: No feedback record is created; the student may retry.
Success Case	Student completes and submits a feedback form for a course they attended.
Trigger	Student navigates to the Feedback section and selects a course to review.
Sequence of Steps	1. Student navigates to the Course Feedback module. 2. System retrieves the list of courses the student is enrolled in that have an open feedback window. 3. Student selects a course from the list. 4. System displays the feedback form (rating options, open-text sections). 5. Student completes all required fields and clicks 'Submit'. 6. System stores the feedback record anonymously (student identity is decoupled from response). 7. System marks this student-course pair as 'feedback submitted'. 8. System displays a confirmation message to the student.
Extensions	• 2a. No courses with open feedback windows: System displays 'No courses available for feedback at this time'. • 5a. Required rating fields are empty: System highlights missing fields; submission is blocked. • 5b. Feedback window has closed since the form was loaded: System displays 'Submission window has closed' and discards the entry. • 7a. Student attempts to re-submit: System detects the duplicate and displays 'You have already submitted feedback for this course'.
Technology	• HTTPS (TLS 1.2+).

	• REST API with anonymisation layer (student ID is replaced with a one-way hash before storage). • Relational database for feedback records. • Submission-deduplication check at the application layer.

UC09 View Marks	
Environment	Flex 2.0 web portal, Academic Records module. Read-only view accessible to the enrolled student only. Available year-round once marks are published.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — monitors academic performance and identifies areas for improvement. • Teacher — publishes marks via Manage Marks Records (UC16); indirectly involved. • Academic Office — maintains authoritative records used for transcripts and progression.
Precondition	• Student is logged in. • The student is or has been enrolled in at least one course. • Marks have been entered and published by the teacher.
Guarantee	• Success: Student's marks are displayed, organised by course and assessment. No data is modified. • Failure: An error message is shown; no data is modified.
Success Case	Student views their assessment results across all enrolled courses.
Trigger	Student clicks 'View Marks' from the academic records menu.

Sequence of Steps	1. Student navigates to the Marks section. 2. System retrieves the student's mark records from the academic database. 3. System organises and displays marks by: Semester > Course > Assessment Type (Quiz, Midterm, Final, Assignment). 4. System calculates and displays the running GPA / average where applicable. 5. Student reviews their results.
Extensions	• 2a. No marks have been published yet: System displays 'No marks available. Check back after results are released'. • 2b. Database error: System displays a generic retrieval error; IT is notified. • 4a. A mark appears incorrect: System shows a link to 'Request Change' (UC11).
Technology	• HTTPS (TLS 1.2+). • REST API GET endpoint with student-scoped data access (RBAC). • Relational database — marks table with foreign keys to students, courses, and assessments.

UC10 View Attendance	
Environment	Flex 2.0 web portal, Attendance module. Read-only view available to the enrolled student. Attendance records are visible once entered by the teacher.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — monitors attendance to avoid academic penalties or eligibility issues.

	• Teacher — records attendance via Manage Attendance Records (UC17); indirectly involved. • Admin — uses attendance data to enforce the minimum attendance policy.
Precondition	• Student is logged in. • Student is enrolled in at least one course. • At least one attendance record exists for the student.
Guarantee	• Success: Attendance records are displayed by course and date. No data is modified. • Failure: An error message is shown; no data is modified.
Success Case	Student views their session-by-session attendance records along with their overall attendance percentage per course.
Trigger	Student clicks 'View Attendance' from the academic records menu.
Sequence of Steps	1. Student navigates to the Attendance module. 2. System retrieves attendance records for all of the student's enrolled courses. 3. System displays attendance per course showing: Date, Session, and Status (Present / Absent / Late). 4. System calculates and displays the overall attendance percentage per course. 5. System highlights any course where the student's attendance is below the required threshold (< 80%).
Extensions	• 2a. No attendance records exist yet: System displays 'No attendance data available'. • 2b. Database error: System displays an error; IT is notified. • 4a. Student believes a session is marked incorrectly: System displays a 'Request Change' link (UC11).
Technology	• HTTPS (TLS 1.2+). • REST API GET endpoint with student-scoped data access.

	• Relational database — attendance table with foreign keys to students, courses, and sessions. • Threshold alert logic at the API or UI layer.

UC11 Request Change	
Environment	Flex 2.0 web portal, Change Request module. Accessible to logged-in students who identify discrepancies in their marks or attendance records.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — wants an error in their academic record corrected. • Admin / Secretariat — receives and processes the request (UC18). • Teacher — may be consulted to verify or dispute the claim. • Academic Office — maintains accurate official records.
Precondition	• Student is logged in. • The student has identified a specific discrepancy in their marks or attendance. • The change request window is open (within the allowed period after results publication).
Guarantee	• Success: A change request record is created and queued for processing by Admin. • Failure: No request record is created; the student is informed of the error.
Success Case	Student successfully submits a documented change request for an incorrect academic record.
Trigger	Student identifies a discrepancy in marks or attendance and clicks 'Request Change'.
Sequence of Steps	1. Student navigates to the Change Request form (accessible from Marks or Attendance views). 2. System pre-fills the affected course and record type based on context. 3. System displays the form: request type (Mark / Attendance), description of the discrepancy, and supporting evidence upload.

	4. Student describes the issue and uploads any supporting documents (screenshot, graded paper). 5. Student submits the request. 6. System validates that all required fields are complete. 7. System creates a change request record with status 'Pending'. 8. System sends a notification to the responsible Admin/Secretariat. 9. System sends an acknowledgement email to the student with a reference number.
Extensions	• 6a. Required fields are empty: System highlights missing fields and blocks submission. • 4a. File upload exceeds size limit or is an invalid format: System shows an upload error. • 5a. Submission window has closed: System displays a message that the request period has ended.
Technology	• HTTPS (TLS 1.2+). • REST API POST endpoint. • File storage system for uploaded documents (max 5 MB, PDF/JPEG/PNG). • SMTP email for student acknowledgement and admin notification. • Relational database for change request records with status tracking.

UC12 Request Maintenance	
Environment	Flex 2.0 web portal, Facilities & Maintenance module. Accessible to all authenticated users to report facility or equipment issues on campus.
Level	*User Goal*
Primary Actor	Teacher

Stakeholders	• Student / Teacher / Staff — reports a campus facility or equipment issue (mostly projector) for resolution. • Saqib Bhai — receives, tracks, and updates maintenance requests (UC25, UC26). • University Administration — monitors campus conditions and resolution rates.
Precondition	• User is logged in. • A physical facility or equipment issue exists on campus.
Guarantee	• Success: A maintenance request is created, logged, and the responsible team is notified. • Failure: No record is created; the user is informed.
Success Case	User submits a detailed maintenance report, which is received by the Saqib Bhai for action.
Trigger	User encounters a campus issue (e.g., broken equipment, facility damage) and navigates to the Maintenance Request form.
Sequence of Steps	1. User navigates to the Maintenance Request form. 2. System displays the form: location (building/room), issue category (Projector, AC, Other), description, urgency level (Low / Medium / High). 3. User fills in all required details. 4. User optionally attaches a photo of the issue (JPEG/PNG, max 5 MB). 5. User submits the request. 6. System validates the form data. 7. System creates a maintenance request record with status 'Open' and assigns a reference number. 8. System notifies Saqib Bhai via the internal dashboard and email. 9. System sends a confirmation to the submitting user with the reference number.
Extensions	• 6a. Required fields are incomplete: System highlights missing fields; submission is blocked.

	• 4a. Photo upload fails (format or size): System displays an upload error; user may submit without a photo. • 8a. Saqib Bhai notification fails: Request record is still saved; IT is alerted to the delivery failure.
Technology	• HTTPS (TLS 1.2+). • REST API POST endpoint. • File storage for photo attachments. • SMTP email notifications. • Relational database for maintenance request records.

UC13 Course Withdraw	
Environment	Flex 2.0 web portal, Enrollment Management module. Available only during the official course withdrawal window defined in the university calendar.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — wishes to drop a course before the deadline to avoid academic penalty. • Academic Office — must update official enrollment records. • Finance Department — may need to process a partial refund depending on the withdrawal date. • Teacher — the course roster must be updated.
Precondition	• Student is logged in. • Student is currently enrolled in the target course. • The course withdrawal window is open. • The student has no administrative holds that block withdrawals.

Guarantee	• Success: Enrollment record is removed; fee record is adjusted; all parties notified. • Failure: Enrollment remains unchanged.
Success Case	Student successfully withdraws from a course after passing eligibility verification, and all downstream records are updated.
Trigger	Student navigates to their enrolled courses and selects 'Withdraw' for a specific course.
Sequence of Steps	1. Student navigates to the Enrolled Courses section. 2. Student selects the course to withdraw from and clicks 'Withdraw'. 3. System displays a confirmation screen showing: potential academic consequences, any applicable refund policy, and a warning if the withdrawal triggers minimum credit-hour issues. 4. System invokes Verification (UC23) to confirm the student meets withdrawal eligibility criteria. 5. Verification returns PASS. 6. Student reads the consequences and confirms withdrawal. 7. System removes the enrollment record from the database. 8. System updates the fee record (applies refund or adjustment as per policy). 9. System notifies the teacher and Academic Office of the change. 10. System sends a confirmation email to the student.
Extensions	• 4a. Verification fails (window closed, account on hold): System displays the specific reason; withdrawal is aborted. • 6a. Student cancels on the confirmation screen: No changes are made. • 8a. Fee adjustment fails: Enrollment is still withdrawn; Finance is alerted to process manually.
Technology	• HTTPS (TLS 1.2+). • REST API with database transactions. • Business rules engine for refund policy calculations.

	• SMTP email notifications.

UC14 Exam Retake Request	
Environment	Flex 2.0 web portal, Examination module. Accessible to students who have failed an exam and qualify for a retake under university policy.
Level	*User Goal*
Primary Actor	Student
Stakeholders	• Student — seeks an opportunity to retake a failed exam to improve their grade. • Admin — processes and approves or rejects retake requests (UC20). • Teacher — will conduct the retake examination if approved. • Examinations Office — schedules the retake and maintains records.
Precondition	• Student is logged in. • The student has a failing grade in an exam eligible for retake under policy. • The retake request window is open.
Guarantee	• Success: A retake request is submitted with verified documents and queued for Admin review. • Failure: No request is created.
Success Case	Student submits a valid exam retake request with supporting documentation, which is forwarded to the Admin for processing.
Trigger	Student navigates to the Exam Retake Request section and selects an eligible exam.
Sequence of Steps	1. Student navigates to the Exam Retake section. 2. System displays a list of exams that are eligible for retake based on the student's academic record. 3. Student selects the exam and fills in the request form (reason for retake, preferred timing). 4. Student uploads required supporting documents (medical certificate, extenuating circumstances letter).

	5. System invokes Verify Documents (UC24) to validate the uploaded documents. 6. Verify Documents returns PASS. 7. Student submits the request. 8. System creates a retake request record with status 'Pending'. 9. System notifies the Admin via the internal dashboard and email. 10. System sends an acknowledgement to the student with a reference number.
Extensions	• 2a. No exams are eligible for retake: System displays 'No exams available for retake at this time'. • 5a. Document verification fails: System displays specific reasons (invalid format, missing signature); student must re-upload. • 4a. File upload error: System prompts the student to retry; submission is blocked until documents are uploaded.
Technology	• HTTPS (TLS 1.2+). • REST API POST endpoint. • Document management system (PDF/JPEG/PNG, max 10 MB). • Document verification service (UC24). • SMTP email notifications.

UC15 Register Users	
Environment	Flex 2.0 Administration panel. Restricted to Admin users only. Used during onboarding of new students, faculty, or staff.
Level	*User Goal*
Primary Actor	Admin
Stakeholders	• Admin — creates and manages user accounts for all system participants.

	• New User (Student / Teacher / Staff) — gains access credentials to Flex 2.0. • IT Security — ensures accounts are created with appropriate roles, permissions, and credential hygiene. • University HR / Admissions — provides the data used to create accounts.
Precondition	• Admin is logged in. • New user's data (name, ID, email, role, department) has been provided and validated offline.
Guarantee	• Success: A new user account is created with the correct role and credentials; the user is notified. • Failure: No account is created; Admin is shown a specific error.
Success Case	Admin successfully registers a new user, who receives their login credentials and can access the system.
Trigger	Admin navigates to User Management and clicks 'Register New User'.
Sequence of Steps	1. Admin navigates to the User Management module. 2. System displays the user registration form: Full Name, University ID, Email, Role (Student/Teacher/Admin/Secretariat), Department, and Contact Number. 3. Admin fills in all required fields. 4. Admin submits the form. 5. System validates that the University ID and email are unique in the database. 6. System creates the user account with a hashed temporary password. 7. System assigns the role and associated permissions (RBAC). 8. System sends a welcome email to the new user containing login credentials and a password-change link. 9. System logs the registration action in the admin audit trail.
Extensions	• 5a. Email already registered: System displays 'Email address is already in use'.

	• 5b. University ID already exists: System displays 'University ID is already registered'. • 8a. Email delivery fails: Account is still created; Admin is shown the temporary credentials to share via alternative means. • 3a. Admin uploads a CSV for bulk registration: System processes each row, reports successes and failures in a summary.
Technology	• HTTPS (TLS 1.2+). • Admin-restricted REST API POST endpoint. • bcrypt password hashing. • RBAC permission assignment at account creation. • SMTP email service. • Audit logging (action, actor, timestamp, IP address).

UC16 Manage Marks Records	
Environment	Flex 2.0 web portal, Academic Records module — Teacher view. Accessible to Teachers for their assigned courses during and after the teaching period.
Level	*User Goal*
Primary Actor	Teacher
Stakeholders	• Teacher — enters and maintains assessment results for their students. • Student — grades directly affect GPA, progression, and scholarship eligibility. • Academic Office — relies on these records for official transcripts and progression decisions. • Admin — oversees data integrity and handles amendment requests from teachers.

Precondition	• Teacher is logged in. • Teacher is assigned to the course in the system. • At least one student is enrolled in the course. • An assessment (Quiz, Assignment, Midterm, or Final) has taken place.
Guarantee	• Success: Marks are saved in the database and become visible to students. • Failure: No changes are made to the database.
Success Case	Teacher enters or updates marks for all students in a course, and the records are saved and published.
Trigger	Teacher navigates to the Marks Management section after an assessment.
Sequence of Steps	1. Teacher navigates to the Marks Management module. 2. System displays the teacher's assigned courses. 3. Teacher selects a course. 4. System displays the list of assessments for that course. 5. Teacher selects an assessment. 6. System displays the list of enrolled students with an input field for each. 7. Teacher enters or updates the mark for each student. 8. Teacher clicks 'Save & Publish'. 9. System validates all entered marks (within the allowed range) 10. System saves the marks to the database. 11. System sends a notification to enrolled students that their marks are available.
Extensions	• 9a. A mark is outside the valid range: System flags the invalid entry; submission is blocked. • 8a. Marks for this assessment were already published: System requires the teacher to submit an amendment request (to prevent unauthorised changes after publishing).

	• 11a. Student notification email fails: Marks are still saved; failure is logged.
Technology	• HTTPS (TLS 1.2+). • REST API PUT/POST endpoint with teacher-scoped authorization. • Relational database with optimistic locking to prevent concurrent edits. • SMTP email for student notifications. • Audit log for all mark changes (previous value, new value, actor, timestamp).

UC17 Manage Attendance Records	
Environment	Flex 2.0 web portal, Attendance module — Teacher view. Available to teachers during and after scheduled class sessions.
Level	*User Goal*
Primary Actor	Teacher
Stakeholders	• Teacher — records which students attended each class session. • Student — attendance records affect eligibility to sit for exams and compliance with university policy. • Admin — enforces minimum attendance rules and generates attendance reports. • Academic Office — attendance records form part of the official student file.
Precondition	• Teacher is logged in. • A scheduled class session exists for the teacher's course. • Students are enrolled in the course.
Guarantee	• Success: Attendance records for the session are saved; students are marked individually.

	• Failure: No changes are saved; prior attendance data is unchanged.
Success Case	Teacher marks the attendance of all students for a specific class session and the data is saved.
Trigger	Teacher navigates to Attendance Management and selects a session to mark.
Sequence of Steps	1. Teacher navigates to the Attendance Management module. 2. System displays the teacher's scheduled sessions for the current and past days. 3. Teacher selects the specific session/date. 4. System displays the complete list of enrolled students. 5. Teacher marks each student as Present, Absent, or Late. 6. Teacher clicks 'Submit Attendance'. 7. System saves the attendance records to the database. 8. System recalculates and updates the running attendance percentage for each student. 9. System flags any student whose attendance has dropped below the required threshold.
Extensions	• 4a. A student appears in the class but is not on the roster: Teacher contacts Admin to resolve the enrollment discrepancy. • 7a. Attendance for this session was already submitted: System requires confirmation before overwriting; change is logged in the audit trail. • 9a. Student falls below attendance threshold: System automatically generates an alert visible to the student and Admin.
Technology	• HTTPS (TLS 1.2+). • REST API PUT/POST endpoint. • Relational database with session-based attendance records. • Real-time attendance percentage calculation. • Audit logging for all changes.

UC18 Process Change Request	
Environment	Flex 2.0 Administration panel, Change Request Management module. Accessible to Admin and Secretariat users. Triggered when a student submits a change request (UC11).
Level	*User Goal*
Primary Actor	Teacher
Stakeholders	• Student — awaits a decision on the accuracy of their academic record. • Teacher — reviews evidence and approves or rejects the request. • Academic Office — affected record must be corrected in the official files if approved.
Precondition	• Teacher is logged in. • At least one change request with status 'Pending' exists in the queue.
Guarantee	• Success: Request is resolved (Approved or Rejected); the academic record is updated if approved; the student is notified. • Failure: Request status remains 'Pending'; Admin is shown an error.
Success Case	Teacher reviews a change request, approves it, and the underlying academic record is corrected.
Trigger	Teacher receives a notification of a pending change request or checks the Change Request queue.
Sequence of Steps	1. Teacher navigates to the Change Request Management module. 2. System displays a list of pending change requests, sorted by submission date. 3. Teacher selects a specific request. 4. System displays full request details: student information, affected record, description of discrepancy, and uploaded documents. 5. Teacher reviews the evidence and may consult the relevant teacher for verification. 6. Teacher selects 'Approve' or 'Reject' and enters a decision note.

	7. If Approved: System updates the relevant marks or attendance record in the database. 8. System changes request status to 'Resolved'. 9. System sends a notification to the student with the decision and admin note.
Extensions	• 5a. Teacher requires more information: Admin sends a query to the student through the system; request status changes to 'Under Review'. • 7a. The underlying record update fails: Teacher is alerted; request remains 'Approved (Pending Update)' until manually resolved. • 6a. Teacher rejects the request: Record is unchanged; student is notified with the reason for rejection.
Technology	• HTTPS (TLS 1.2+). • Admin-restricted REST API. • Relational database for change request records and academic records. • SMTP email notifications. • Audit logging (decision, actor, timestamp).

UC19 Process Extra Class Request	
Environment	Flex 2.0 Secretariat panel, Scheduling & Extra Classes module. Accessible to Secretariat users. Triggered when a teacher submits an extra class request (UC21).
Level	*User Goal*
Primary Actor	Secretariat
Stakeholders	• Teacher — awaits approval and room assignment for the extra session. • Student — needs to be notified of the additional class. • Secretariat — validates availability and approves/rejects the request.

	• Facilities Management — provides room and resource allocation.
Precondition	• Secretariat is logged in. • A pending extra class request exists (submitted via UC21).
Guarantee	• Success: Request is approved, room assigned, schedule updated, and all parties notified. • Failure: Request is rejected with reason; teacher is informed.
Success Case	Secretariat reviews the request, confirms room availability, approves the extra class, and notifies teacher and students.
Trigger	Secretariat receives a notification of a pending extra class request.
Sequence of Steps	1. Secretariat navigates to the Extra Class Request queue. 2. System displays pending requests with proposed subject, time, and number of students. 3. Secretariat selects a request to process. 4. System invokes Check Availability (UC22) for the proposed time slot and resource requirements. 5. Check Availability returns an available room. 6. Secretariat reviews the result and assigns the confirmed room to the request. 7. Secretariat approves the request. 8. System creates an extra class session record in the schedule. 9. System notifies the teacher with the confirmed time and room. 10. System notifies all enrolled students of the additional class session.
Extensions	• 5a. No rooms are available at the proposed time: System displays alternative available slots; Secretariat proposes an alternative to the teacher. • 5b. Check Availability returns an error: Secretariat manually checks the room booking register and updates the request. • 7a. Secretariat rejects the request: System notifies the teacher with the reason.

Technology	• HTTPS (TLS 1.2+). • REST API with scheduling module integration. • Room booking / scheduling database. • SMTP email notifications to teacher and all enrolled students. • Audit logging.

UC20 Process Exam Retake Request	
Environment	Flex 2.0 Administration panel, Examinations module. Accessible to Admin users. Triggered when a student submits a verified exam retake request (UC14).
Level	*User Goal*
Primary Actor	Admin
Stakeholders	• Student — awaits a decision on whether they will be allowed to retake the exam. • Admin — reviews eligibility criteria and supporting documents. • Teacher — will check the exam if the retake happens. • Examinations Office — schedules and records the retake exam.
Precondition	• Admin is logged in. • At least one exam retake request with status 'Pending' exists, with verified documents.
Guarantee	• Success: Retake is approved, exam is scheduled, and all parties are notified. • Failure: Retake is rejected; student is notified with the reason.
Success Case	Admin reviews a retake request, confirms the student's eligibility, approves the retake, and schedules the exam.
Trigger	Admin receives a notification of a pending exam retake request or reviews the Examinations queue.
Sequence of Steps	1. Admin navigates to the Exam Retake Request queue. 2. System displays pending retake requests with student name, failed exam, and submission date. 3. Admin selects a request. 4. System displays full details: student's academic history, failed grade, submitted documents, and request reason. 5. Admin reviews the student's eligibility against retake policy.

	6. Admin approves or rejects the request, entering a decision note. 7. If Approved: System schedules the retake exam in the examination timetable. 8. System updates the request status to 'Approved' or 'Rejected'. 9. System notifies the student and relevant teacher of the decision and (if approved) the scheduled date.
Extensions	• 5a. Documents are insufficient or suspicious: Admin rejects with reason 'Insufficient documentation'. • 7a. A scheduling conflict exists: Admin manually selects an alternative date and time; system updates the schedule. • 6a. Student has exceeded the maximum number of retakes: System auto-flags the request; Admin cannot approve without an override.
Technology	• HTTPS (TLS 1.2+). • Admin-restricted REST API. • Examination scheduling database. • Document management system (view uploaded documents). • SMTP email notifications. • Audit logging.

UC21 Request Extra Class	
Environment	Flex 2.0 web portal, Scheduling module — Teacher view. Accessible to Teachers when they need to arrange additional instructional sessions beyond the standard timetable.
Level	*User Goal*
Primary Actor	Teacher

Stakeholders	• Teacher — needs an additional class session to cover curriculum gaps or missed content. • Student — benefits from additional instruction; must be informed in advance. • Secretariat — approves the request and assigns a room (UC19).
Precondition	• Teacher is logged in. • Teacher is assigned to at least one course. • An academic need for an extra session exists.
Guarantee	• Success: An extra class request is submitted and the Secretariat is notified. • Failure: No request is created; Teacher is informed of the error.
Success Case	Teacher submits an extra class request with a proposed time, subject, and reason, and the Secretariat receives it for processing.
Trigger	Teacher navigates to the Extra Class Request form and submits a request.
Sequence of Steps	1. Teacher navigates to the Extra Class Request module. 2. System displays the request form: Course, Topic to be covered, Proposed Date and Time, Estimated Duration, and Reason for extra class. 3. Teacher completes all required fields. 4. System warns the teacher if the proposed time overlaps with any of their existing scheduled sessions. 5. Teacher submits the request. 6. System validates the form data. 7. System creates the extra class request with status 'Pending Approval'. 8. System sends a notification to the Secretariat. 9. System sends an acknowledgement to the teacher with a reference number.
Extensions	• 4a. Proposed time overlaps with teacher's own schedule: System warns but allows the teacher to proceed or change the time.

	• 6a. Required fields are empty: System highlights missing fields; submission is blocked. • 8a. Secretariat notification fails: Request is still saved; failure is logged.
Technology	• HTTPS (TLS 1.2+). • REST API POST endpoint. • Schedule conflict-detection logic. • SMTP email notifications. • Relational database for request records.

UC22 Check Availability	
Environment	Flex 2.0 back-end service layer, Scheduling module. An internal sub-function invoked automatically when the Secretariat processes an extra class request. Not directly accessible to users.
Level	*Sub-function*
Primary Actor	System (invoked by Process Extra Class Request — UC19)
Stakeholders	• Secretariat — needs accurate room and time-slot availability data to make a decision. • Facilities Management — room assignments must be conflict-free. • Teacher and Students — rely on the confirmed booking being accurate.
Precondition	• Triggered as part of Process Extra Class Request (UC19). • A proposed date, time, and resource requirement are available.
Guarantee	• Success: Returns an available room and time slot, or a list of suggested alternatives. • Failure: Returns 'Unavailable' with a reason, or an error to the calling use case.
Success Case	System confirms that the proposed time slot and a suitable room are available and returns the details.
Trigger	Invoked via «include» by Process Extra Class Request (UC19).
Sequence of Steps	1. System receives the proposed date, time range, and capacity requirement. 2. System queries the room booking database for all rooms with sufficient capacity. 3. System checks each candidate room for schedule conflicts in the specified time window. 4. System checks that the teacher has no conflicting session at the proposed time.

	5. System returns the first available room and time slot, along with up to three alternative slots if available.
Extensions	• 3a. All suitable rooms are booked: System returns 'No availability at proposed time' with the nearest available alternative slots. • 2a. Room booking database is unreachable: System returns an error to UC19; Secretariat is prompted to check manually.
Technology	• Internal REST API or database stored procedure. • Room booking / scheduling relational database. • Conflict-detection algorithm (interval overlap check).

UC23 Verification	
Environment	Flex 2.0 back-end service layer, Enrollment module. An internal sub-function invoked automatically when a student initiates a Course Withdraw. Not directly accessible to users.
Level	*Sub-function*
Primary Actor	System (invoked by Course Withdraw — UC13)
Stakeholders	• Student — must satisfy eligibility conditions before withdrawal is processed. • Academic Office — policy rules govern when withdrawals are permissible. • Finance Department — a withdrawal that bypasses verification could affect fee records incorrectly.
Precondition	• Triggered as part of Course Withdraw (UC13). • Student ID, Course ID, and current date are available.
Guarantee	• Success: Returns PASS — the student meets all withdrawal eligibility criteria. • Failure: Returns FAIL — with a specific reason (window closed, account hold, minimum credits violated).
Success Case	System confirms that the student is eligible to withdraw from the course and returns PASS to the calling use case.
Trigger	Invoked via «include» by Course Withdraw (UC13) immediately after student confirmation intent.
Sequence of Steps	1. System retrieves the current date and checks whether the course withdrawal window is still open. 2. System checks the student's account for any administrative or financial holds. 3. System calculates the student's remaining credit hours if the withdrawal is processed.

	4. System checks whether the resulting credit load meets the minimum required by university policy. 5. System returns PASS or FAIL with a specific reason to UC13.
Extensions	• 1a. Withdrawal window is closed: Immediately returns FAIL — 'Withdrawal period has ended'. • 2a. Student has a financial hold: Returns FAIL — 'Account hold: please contact Finance'. • 4a. Withdrawal would drop student below minimum credits: Returns FAIL — 'Insufficient credit hours remaining'.
Technology	• Internal REST API or database stored procedure. • Relational database — enrollment, student account, and academic calendar tables.

UC24 Verify Documents	
Environment	Flex 2.0 back-end service layer, Document Management module. An internal sub-function invoked during Exam Retake Request processing. May involve automated checks and, for flagged documents, a manual Admin review step.
Level	*Sub-function*
Primary Actor	System (invoked by Exam Retake Request — UC14); Admin for manual review
Stakeholders	• Student — submitted documents must be valid for the request to proceed. • Admin — reviews flagged documents that fail automated checks. • Examinations Office — only verified, authentic documentation should support a retake claim.
Precondition	• Triggered as part of Exam Retake Request (UC14). • The student has uploaded at least one supporting document.

Guarantee	• Success: Returns PASS — all uploaded documents are verified as valid and authentic. • Failure: Returns FAIL — with a specific reason (wrong type, unreadable, suspected tampering).
Success Case	System validates all uploaded documents and returns PASS to the Exam Retake Request use case.
Trigger	Invoked via «include» by Exam Retake Request (UC14) when the student uploads supporting documents.
Sequence of Steps	1. System receives the uploaded document files from the student's submission. 2. System checks each document's file type (must be PDF, JPEG, or PNG) and size (max 10 MB). 3. System performs metadata validation to detect obvious tampering (e.g., altered creation date, mismatched fonts in PDF). 4. System checks that the document is not a duplicate of a previously submitted document. 5. If automated checks pass: System returns PASS to UC14. 6. If a document is flagged: System routes it to the Admin review queue and pauses the request. 7. Admin manually reviews flagged document and overrides with PASS or FAIL.
Extensions	• 2a. File type is not supported: Returns FAIL — 'Unsupported file format. Please upload PDF, JPEG, or PNG'. • 2b. File size exceeds limit: Returns FAIL — 'File size exceeds the 10 MB limit'. • 3a. Document appears tampered: Flagged for Admin review; request is paused. • 4a. Duplicate document detected: Returns FAIL — 'This document has already been submitted'.
Technology	• File type validation (MIME type inspection). • PDF metadata inspection library.

	• Secure file storage with hash-based deduplication. • Admin review portal within Flex 2.0. • Audit logging of all verification decisions.

UC25 View Maintenance Request	
Environment	Flex 2.0 Secretariat / Admin panel, Facilities & Maintenance module. Accessible to Saqib Bhai to monitor all submitted maintenance requests.
Level	*User Goal*
Primary Actor	Saqib Bhai
Stakeholders	• Admin — monitors campus maintenance status and uses data for facilities planning. • Submitting User — can track the status of their own request. • Saqib Bhai— receive task assignments through this view.
Precondition	• Saqib Bhai is logged in. • At least one maintenance request exists in the system.
Guarantee	• Success: Maintenance request details are displayed; no data is modified. • Failure: An error message is shown; no data is modified.
Success Case	Saqib Bhai views a list of all maintenance requests, filters them by status, and opens a specific request to review its full details.
Trigger	Saqib Bhai navigates to the Maintenance Management module to review outstanding requests.
Sequence of Steps	1. Saqib Bhai navigates to the Maintenance Request Management module.

	2. System retrieves and displays all maintenance requests with: Reference Number, Category, Location, Urgency Level, Submission Date, and Status (Open, In Progress, Resolved, Closed). 3. Saqib Bhai applies filters (by status, location, date range, or urgency level). 4. Saqib Bhai selects a specific request. 5. System displays full details: submitter information, issue description, attached photos, and status history.
Extensions	• 2a. No maintenance requests exist: System displays 'No maintenance requests found'. • 3a. No results match the applied filter: System displays 'No requests match the selected filters'. • 5a. Attached photo or document is unavailable: System shows 'Attachment unavailable' with an error code.
Technology	• HTTPS (TLS 1.2+). • REST API GET endpoint with role-scoped access (Secretariat/Admin only). • Relational database with maintenance request status history. • File storage for photo attachments. • Pagination and server-side filtering.

UC26 Update Maintenance Request	
Environment	Flex 2.0 Saqib Bhai panel, Facilities & Maintenance module. Accessed after reviewing a maintenance request (UC25) to update its status and communicate progress to the submitter.
Level	*User Goal*
Primary Actor	Saqib Bhai

Stakeholders	• Saqib Bhai — updates request status as maintenance work progresses. • Facilities / Maintenance Staff — status changes communicate next steps. • Submitting User — receives notifications on the progress and resolution of their reported issue. • Admin — monitors resolution rates and outstanding issues.
Precondition	• Saqib Bhai is logged in. • A maintenance request exists with a status other than 'Closed'.
Guarantee	• Success: Maintenance request status is updated, a status note is saved, and the submitter is notified. • Failure: No changes are made to the request record.
Success Case	Saqib Bhai updates the status of a maintenance request to 'Resolved', and the original submitter is notified.
Trigger	Saqib Bhai opens a maintenance request in View Maintenance Request (UC25) and clicks 'Update Status'.
Sequence of Steps	1. Saqib Bhai navigates to a specific maintenance request (via UC25). 2. System displays the current status and a status update form: new status selection, update note, and estimated completion date (if applicable). 3. Saqib bhai selects the new status (In Progress / Resolved / Closed) and enters an update note. 4. Saqib Bhai clicks 'Submit Update'. 5. System validates that the status transition is valid (e.g., cannot move from 'Closed' back to 'Open'). 6. System saves the updated status and note, and appends an entry to the request's status history. 7. System sends an email notification to the original submitter with the new status and note.
Extensions	• 5a. Invalid status transition (e.g., 'Closed' → 'Open'): System rejects and displays an error; current status is unchanged.

	• 3a. Note field is empty: System requires a minimum note for status changes to 'Resolved' or 'Closed'. • 7a. Email notification fails: Status update is still saved; failure is logged for IT to investigate.
Technology	• HTTPS (TLS 1.2+). • REST API PATCH endpoint. • Relational database with append-only status history log. • State machine validation for status transitions. • SMTP email notifications. • Audit logging (actor, old status, new status, timestamp).